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ASCEND 2030

Getting Ready:
How Computers Hold Data



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UTM BIG DATA CENTRE
TOGETHER WE MAKE MOTIONS...

Menginovasi Penyelesaian Lestari

DFBOOK DIGITAL FORENSICS · LESSON 0.0

Getting Ready: How Computers Hold Data

Lesson overview: the basics every examiner relies on



LESSON 0.0 · OVERVIEW

Getting Ready

Digital evidence is data, and data has its own language

BY THE END YOU WILL BE ABLE TO

Read bits, bytes and hexadecimal

How a computer writes data, and how examiners read it

Tell a file from its storage

What a file is, and where its data really lives

Tell what is lost at power-off

Volatile memory versus persistent storage

Find your way around a forensics tool

Open a case in Autopsy and move between its panes

LESSON 0.0 · ROUTE

Three Short Sub-Lessons

One idea and one short activity in each

LESSON 0.1

Bits, Bytes and Hexadecimal

How a computer writes data, and how examiners read it.

ACTIVITY **Conversion drill**

LESSON 0.2

Files, Storage and Memory

What a file is, where it lives, and what is lost at power-off.

ACTIVITY **Sorting exercise**

LESSON 0.3

Meeting the Tools

Opening a case in Autopsy and finding your way around.

ACTIVITY **Tool orientation**

Practise only on the course training image, never on a real device. A training image is a forensic image: an exact copy of a whole drive, saved as one file or a numbered set of files.

UP NEXT

Bits, Bytes and Hexadecimal

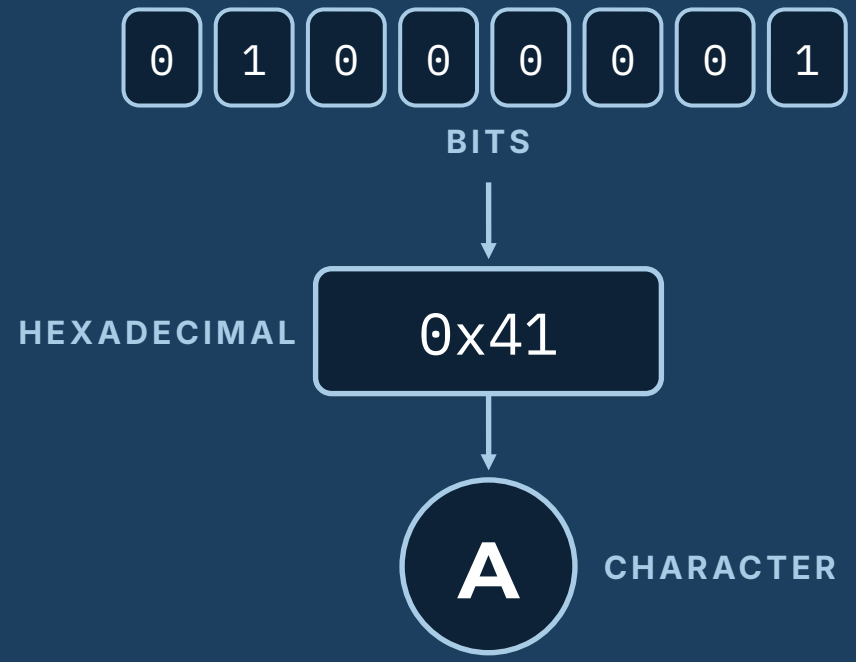
We start with the smallest unit of data, and the number system examiners use to read it.

- **Bits and bytes**
- **Counting in base 16**
- **Reading a byte**

DFBOOK DIGITAL FORENSICS · LESSON 0.1

Bits, Bytes and Hexadecimal

How a computer writes data, and how examiners read it



IN THIS LESSON

1. Bits and bytes
2. Counting in base 16
3. Reading a byte

LESSON 0.1 · OVERVIEW

Bits, Bytes and Hexadecimal

Forensics tools show raw data in hexadecimal, or hex for short. File headers, hash values and deleted-file markers are all written in hex; this lesson ends by showing each one.

BY THE END YOU WILL BE ABLE TO

1. Explain what a bit, a byte and ASCII are
2. Count from 0 to F and write hex values
3. Convert a byte between binary, decimal and hex

INTRODUCTION · WHY THIS MATTERS

Why This Matters

Three numbers sit under every piece of digital evidence

2

States of a bit

Off or on, written 0 or 1. Everything a computer stores is built from bits.

8

Bits in a byte

Bits are grouped in eights. Tools count and display data in bytes.

16

Symbols in hex

0 to 9, then A to F. Hex is how forensics tools show raw data.

WHERE YOU WILL MEET HEX

- Hex views
- File headers
- Hash values
- Deleted files

INTRODUCTION · ROADMAP

Your Route Through Lesson 0.1

Each part answers one question, in order

1

PART 1

Bits and bytes

How does a computer store information?

2

PART 2

Counting in base 16

How is hexadecimal counted and written?

3

PART 3

Reading a byte

How do you turn a byte into decimal and hex, and back?

One case runs through the module: Teratai Supplies, where the training image you meet in Lesson 0.3 comes from.

PART 1 OF 3

Bits and Bytes

The smallest units of data, and how they become characters

01

IN THIS PART YOU WILL LEARN TO

- Describe a bit, a byte and binary
- Explain what ASCII does

KEY TERMS

Bit **Byte** **Binary** **ASCII**

PART 1 · DEFINITION

Bits, Bytes and ASCII

One sentence, three terms

Information in a computer is stored in (1) bits that are off or on, grouped into (2) bytes of eight, and shown as text through (3) ASCII.

1 Bit

One binary digit: 0 (off) or 1 (on). Binary means counting with only 0 and 1 (base 2).

2 Byte

Eight bits read as one unit. Data is stored and counted in bytes.

3 ASCII

American Standard Code for Information Interchange: bit patterns for letters, digits and symbols.

Raw binary is hard to read: in a long run of 0s and 1s, it is hard to see where one byte ends.

PART 1 · MECHANISM

From Switches to a Character

Left to right: eight bits, one byte, one character

1 · EIGHT BITS



Each switch is off (0) or on (1).

2 · ONE BYTE

0100 0001

Eight bits, read as one unit.

3 · ONE CHARACTER

A

Read as text, ASCII says this pattern is the letter A.

Context matters: the same byte can be a letter, a number or part of a picture. ASCII applies when bytes are read as text; Unicode extends it to more languages.

PART 2 OF 3

Counting in Base 16

Why examiners read hex instead of long runs of binary

02

IN THIS PART YOU WILL LEARN TO

- Count from 0 to F and past it
- Write hex values with 0x or h

KEY TERMS

Base 10 **Base 16** **Hex digit** **0x**

PART 2 · COMPARISON

Base 10 and Base 16

Read each row across: the same idea in two number systems

	Base 10 (decimal)	Base 16 (hexadecimal)
Symbols	0 to 9: ten symbols	**0 to 9, then A to F: sixteen symbols**
After the last	9 is followed by 10 (ten)	F is followed by 10 (sixteen)
Used by	People, every day	Examiners reading binary data
Same amount	18	0x12, said “one-two”

Hexadecimal makes the same binary data easier to view.

PART 2 · CONCEPT

Counting from 0 to F

The sixteen hex digits and their decimal values

HEX DIGIT	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
DECIMAL	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15

A to F are digits here, worth 10 to 15

AFTER F, CARRY ONE, AS DECIMAL DOES AFTER 9

```
0x0F  =  15
```

```
0x10  =  16
```

```
0x1F  =  31
```

```
0xFF  =  255
```

Hex digit: one of the sixteen symbols 0 to F. Read hex digit by digit: 0x41 is “four-one”, not forty-one.

PART 2 · NOTATION

Writing Hex Values

Each row: a writing convention on the left, an example on the right

EXAMPLE

The 0x prefix

Written in front of a value to mark it as hexadecimal.

`0x09`

The h suffix

Some programs and documents write h after the value instead.

`Bh  (B = 11)`

Two digits per byte

A byte is always two hex digits, so 9 is 09 and B is 0B.

`0x09 · 0x0B`

0x12: hex one-two, which is 18 in decimal. A plain 12 is twelve: the same digits can mean different amounts, so the prefix matters.

PART 3 OF 3

Reading a Byte

Turning a byte into decimal and hex, and back again

03

IN THIS PART YOU WILL LEARN TO

- Use position values, read right to left
- Convert a byte to decimal and hex, and back

KEY TERMS

Position value **Hex digit** **Hex view**

PART 3 · METHOD

Bit Position Values

Each of the eight positions has a fixed value; read from right to left

POSITION VALUE · HEX	0x80	0x40	0x20	0x10	0x08	0x04	0x02	0x01
BIT	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1
POSITION VALUE · DECIMAL	128	64	32	16	8	4	2	1

FIRST HEX DIGIT · LEFT FOUR BITS

SECOND HEX DIGIT · RIGHT FOUR BITS

READ RIGHT TO LEFT

1. Add the values of every position whose bit is 1.
2. Each hex digit covers four bits, so a byte is always two hex digits.

PART 3 · WORKED EXAMPLE

The Letter A

Left: the byte. Right: three steps to its value

{} THE CASE

One byte from a text file in a hex view:

```
0100 0001
```

QUESTION

What value is it, and which character?

1 **Mark the bits that are on**

The 64 position (0x40) and the 1 position (0x01).

2 **Add the decimal values**

64 + 1 = 65.

3 **Add the hex values**

0x40 + 0x01 = 0x41.

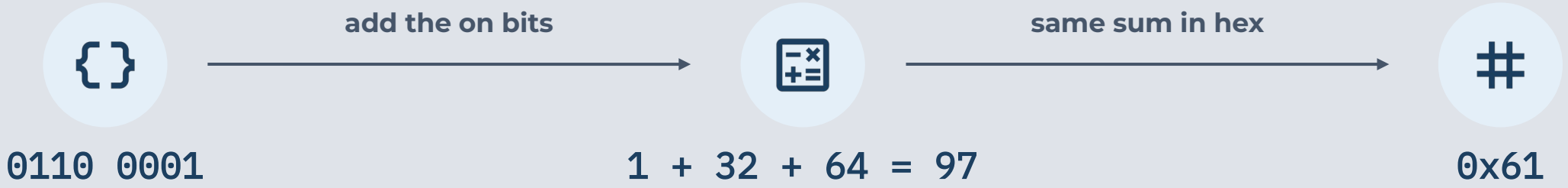
ANSWER

65 in decimal, 0x41 in hex: the ASCII letter A.

PART 3 · WORKED EXAMPLE

The Lowercase Letter a

Left to right: the bits, the decimal sum, the hex value



1 Find the bits that are on

Positions 64, 32 and 1 (0x40, 0x20 and 0x01).

2 Compare with A

One extra bit, worth 32 (0x20), turns A into a.

ANSWER

97 in decimal, 0x61 in hex: the ASCII letter a.

Upper and lower case are different bytes, so a search can match one and miss the other.

PART 3 · WORKED EXAMPLE

A Deleted-File Marker: E5

Left: the byte. Right: from hex back to bits and decimal

THE CASE

In a USB stick's hex view, where a file name starts:

E5

QUESTION

What are its bits and value?

1 Split it into two hex digits

E and 5. Each hex digit stands for four bits.

2 Turn each digit into four bits

E = 14 = 8 + 4 + 2, so 1110. 5 = 4 + 1, so 0101.

3 Add the position values

1110 0101: 128 + 64 + 32 + 4 + 1 = 229.

ANSWER

0xE5 = 1110 0101 = 229

On FAT (File Allocation Table) drives, it marks a deleted file.

PART 3 · STRUCTURE

Reading a Hex View

Two mock rows from a text file: position, bytes and text

Hex view · Invoice_note.txt MOCK

OFFSET 1	BYTES 2															TEXT 3
00000000	**54**	65	72	61	74	61	69	20	49	6E	76	6F	69	63	65	20	Teratai Invoice
00000010	30	34	31	32	0D	0A	54	6F	74	61	6C	3A	20	52	4D	20	0412..Total: RM

1 Offset

Where the row starts, in hex. Row two starts at 0x10: byte 16, counting from 0.

2 Bytes

Sixteen per row, two hex digits each. 0x54 is T.

3 Text

The bytes read as ASCII. A dot marks a byte with no character, such as a line break.

SIZE UNITS **1 KB = 1,024 bytes · 1 MB = 1,024 KB · 1 GB = 1,024 MB · 1 TB = 1,024 GB**

PART 3 · APPLICATION

Where Hex Appears in This Module

Four places you will read hex, and the lesson where each one returns

Hex views

Tools show raw data in hex, 16 bytes a row.

`54 65 72`

LESSON 0.3 · 5.1

Hash values

A fixed-length fingerprint of data, shown in hex; one changed bit changes it.

`9E10 7D9D …`

LESSON 3.4

Deleted files

FAT, a file system, replaces the first letter of a deleted file's name with E5.

`E5`

LESSON 4.3

File headers

JPEG (a photo format) files always start with the bytes FF D8.

`FF D8`

LESSON 5.1

Tools show bytes without 0x: read every pair as hex.

LESSON 0.1 · SUMMARY

Lesson Summary

Three ideas, in the order you met them

1

Bits and bytes

A bit is off or on; a byte is eight bits; read as text, ASCII gives each pattern a character.

2

Base 16

Hex digits run 0 to F, with A to F worth 10 to 15. A hex value is marked 0x or h.

3

Reading a byte

Add the on bits: 0100 0001 = 65 = 0x41. Back again: 0xE5 = 1110 0101 = 229.

Hex is how examiners read raw data.

UP NEXT

Files, Storage and Memory

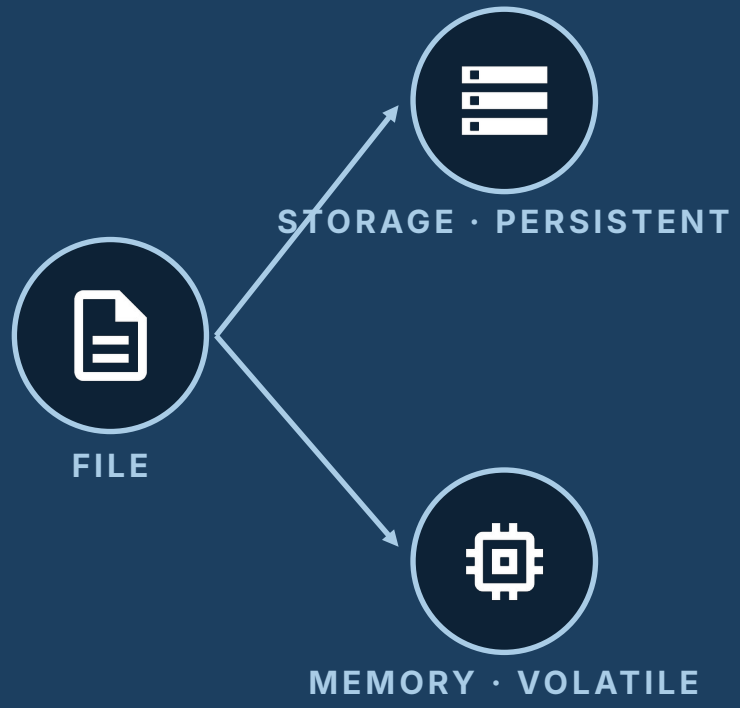
Next, where those bytes live: the difference between a file and its storage, and what disappears at power-off.

- **Files and storage**
- **Inside a storage device**
- **Memory versus storage**

DFBOOK DIGITAL FORENSICS · LESSON 0.2

Files, Storage and Memory

What a file is, where it lives, and what is lost at power-off



IN THIS LESSON

1. Files and storage
2. Inside a storage device
3. Memory versus storage

LESSON 0.2 · OVERVIEW

Files, Storage and Memory

A file you see in a folder is not the same thing as the data on the drive, and data in memory is not the same as data on the drive. Each difference decides what an examiner can still find.

BY THE END YOU WILL BE ABLE TO

1. **Explain what a file system does**
2. **Name the parts of a storage device**
3. **Tell volatile data from persistent data**

INTRODUCTION · WHY THIS MATTERS

Why This Matters

Three places the same evidence can be, with different rules

> **“A file system provides an OS [operating system] with a road map to data on a disk.”**
>
> Guide to Computer Forensics and Investigations, on file systems

The file you see

A name in a folder: an entry in the road map, not the data itself.

The data on the drive

Bits written to the drive. They can remain after the file is deleted.

The data in memory

Exists while the computer runs, and is lost at power-off.

File system: the part of the operating system that records where each file's data is stored on a drive.

INTRODUCTION · ROADMAP

Your Route Through Lesson 0.2

Each part answers one question, in order

1

PART 1

Files and storage

What is the difference between a file and the storage that holds it?

2

PART 2

Inside a storage device

How is data laid out on a drive?

3

PART 3

Memory versus storage

What is lost when a computer is switched off?

One case runs through the module: Teratai Supplies, whose accounts laptop and deleted invoices appear in the examples.

PART 1 OF 3

Files and Storage

What the user sees, and what the drive actually holds

01

IN THIS PART YOU WILL LEARN TO

- Describe what a file system does
- Explain why deleted files can remain on a drive

KEY TERMS

File system | Partition | Cluster | Free space

PART 1 · TWO VIEWS

Files and the Data Behind Them

Left: what the operating system shows. Right: what the drive holds

WHAT THE USER SEES

Files in folders

Names, folders and icons that the operating system shows from its road map.

BUILT FROM

The file system’s entries

WHAT THE DRIVE HOLDS

Bits in sectors

The data itself, written to the drive in fixed-size sections called sectors.

SECTOR SIZE

512 to 4,096 bytes

HOW THEY LINK

File name → File system entry → Sectors on the drive

PART 1 · STRUCTURE

Partitions, File Systems and Clusters

How one drive is divided before any file is saved

ONE DRIVE



Clusters in partition 1. Each holds four sectors. Filled: allocated to a file. Dashed: unallocated.

1 Partition

A section of a drive that works as its own logical drive, such as C:.

2 File system

Each partition has one: NTFS (New Technology File System) on Windows, FAT on USB sticks.

3 Cluster

The unit a file system gives to files: one or more sectors grouped together.

Where to look: deleted data usually sits in unallocated space (also called free space), the clusters no file is using.

PART 1 · WORKED EXAMPLE

The Deleted Invoice

Teratai Supplies: a clerk deletes an invoice. Is it gone?



1 What changed

The entry is marked deleted, so the folder hides it. Its clusters become free space for reuse.

2 What stayed

The data, until new data is written over it. A solid-state drive (SSD) may erase it sooner (TRIM).

ANSWER

Usually not: a forensics tool such as Autopsy (Lesson 0.3) can often recover it.

PART 2 OF 3

Inside a Storage Device

Where the bytes physically sit on a drive

02

IN THIS PART YOU WILL LEARN TO

- Name the parts of a hard disk drive
- Explain how a solid-state drive relates to it

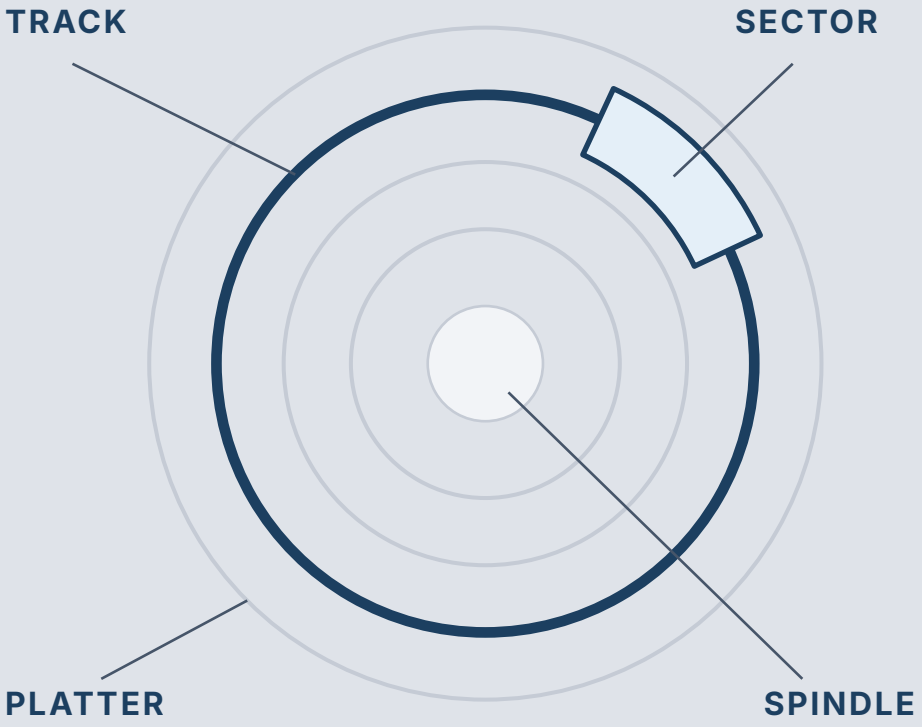
KEY TERMS

Platter **Head** **Track** **Cylinder** **Sector**

PART 2 · STRUCTURE

Parts of a Hard Disk Drive

Left: one platter from above (heads and cylinders are not visible). Right: the five parts



Platter	A disk coated with magnetic material; a drive has one or more.
Head	Reads and writes data; two per platter, one for each side.
Track	One of the rings, one inside another, where data is stored.
Cylinder	The tracks in the same position on stacked platters.
Sector	A section of a track, typically 512 to 4,096 bytes.

PART 2 · COMPARISON

Hard Disks and Solid-State Drives

Left: a hard disk drive. Right: a solid-state drive

Hard disk drive (HDD)

- Platters coated with magnetic material
- Heads that move to read and write
- Data laid out in tracks and sectors

Solid-state drive (SSD)

- Flash memory chips instead of platters
- Shows the computer the same sectors
- **May erase deleted data by itself (TRIM)**

Geometry: how a drive is laid out in platters, tracks and sectors. SSDs copy it, so the same imaging tools work, but deleted data is less likely to survive.

PART 3 OF 3

Memory Versus Storage

Why some evidence disappears when the power goes off

03

IN THIS PART YOU WILL LEARN TO

- Tell volatile data from persistent data
- Explain why a running computer needs care

KEY TERMS

RAM | Volatile | Persistent | Order of volatility

PART 3 · CONCEPT

How Long Data Lasts

From the shortest-lived data on the left to the longest-lived on the right

RAM and running processes

Volatile: lost at power-off.

Files stored on the drive

Persistent: kept for years.

MILLISECONDS → YEARS

Order of volatility: the order for capturing evidence, starting with the data that is lost fastest.

RAM: random access memory, the computer's working memory. Its contents are lost when the computer is switched off.

Any action on a running computer changes its RAM. Lesson 7.1 shows how examiners handle this.

PART 3 · WORKED EXAMPLE

The Laptop Left Running

Left: the scene. Right: what each item would do at power-off

THE CASE

The accounts laptop is found switched on, with a banking session open.

QUESTION

What would be lost if it were switched off now?

1. **Contents of RAM** — VOLATILE
   Working data held only while the laptop runs.
2. **The open banking session** — VOLATILE
   A running program, held in memory.
3. **Invoices saved on the drive** — PERSISTENT
   Written to the drive; they survive power-off.

ANSWER

Power-off loses RAM and the session; saved files remain.

PART 3 · WORKED EXAMPLE

Six Items from the Case

Each item sorted by how long it lasts

THE CASE

Teratai Supplies: six items found on or around the accounts laptop.

VOLATILE	PERSISTENT	DELETED · UNTIL OVERWRITTEN
Contents of RAM	**Saved spreadsheet**	**Deleted invoice**
Open banking session	**Files on the USB stick**	**Deleted-file fragments**
Lost at power-off.	Kept until someone deletes them.	Still on the drive, in free space.

ANSWER

Capture volatile data first. If the drive is encrypted (locked with a key), get advice before switching off.

LESSON 0.2 · SUMMARY

Lesson Summary

Three ideas that lead to one habit

Files and storage

A file is an entry in the road map. Deleted data can stay in free space until overwritten.

Inside a storage device

Platters, heads, tracks, cylinders and sectors. SSDs copy the layout but may erase deleted data.

Memory versus storage

RAM and running processes are volatile; files on a drive are persistent.

Before you touch a device, know what you are looking at: a file, its storage, or memory.

UP NEXT

Meeting the Tools

Next, you open a forensics tool for the first time and learn to find your way around it.

- **Before you open a tool**
- **Opening a case**
- **Finding your way around**

DFBOOK DIGITAL FORENSICS · LESSON 0.3

Meeting the Tools

One rule, one workflow and one screen layout



IN THIS LESSON

1. Before you open a tool
2. Opening a case
3. Finding your way around

LESSON 0.3 · OVERVIEW

Meeting the Tools

A forensics tool shows everything in a forensic image, including what the user deleted. Before you use one, you need one rule, one workflow and one screen layout.

BY THE END YOU WILL BE ABLE TO

1. **Explain why examiners work on a forensic image**
2. **Open a case and add a forensic image in Autopsy**
3. **Find a file, label it (tag) and save a copy (extract)**

INTRODUCTION · WHY THIS MATTERS

Why This Matters

Left: an everyday file browser. Right: a forensics tool

EVERYDAY FILE BROWSER

Shows what the road map lists

- ✓ Existing files in folders
- ✗ Deleted files and fragments: not shown
- ✗ Raw bytes in hexadecimal: not shown

FORENSICS TOOL

Shows what the drive holds

- ✓ Existing files in folders
- ✓ Deleted files and fragments
- ✓ Raw bytes in hexadecimal

Autopsy: a free digital forensics tool built on The Sleuth Kit. It can recover deleted files, and you use it throughout this course.

INTRODUCTION · ROADMAP

Your Route Through Lesson 0.3

Each part answers one question, in order

1

PART 1

Before you open a tool

Why do examiners never work on the original drive?

2

PART 2

Opening a case

How do you load a forensic image into Autopsy?

3

PART 3

Finding your way around

Where do you find a file, and what can you do with it?

One case runs through the module: Teratai Supplies. The course training image you open in Autopsy comes from this case.

PART 1 OF 3

Before You Open a Tool

The rule that protects the evidence

01

IN THIS PART YOU WILL LEARN TO

- State the first rule of digital forensics
- Tell a forensic image from a backup

KEY TERMS

Forensic image **Hash** **Write-blocker**

PART 1 · PRINCIPLE

The First Rule

Preserve the original, then analyse only a copy



Original drive

Secured and kept unchanged.

Forensic image

An exact copy of every bit.

Your analysis

Every search and export is done here.

Forensic image: one file, or a numbered set, with an exact copy of every bit on a drive; also called a bit-stream image.

Hash: a fixed-length fingerprint, made by a method such as MD5 or SHA-1. Matching hashes of original and image show the copy is exact.

Write-blocker: a device that stops any writing to the original while it is copied. Lesson 3 covers it.

PART 1 · COMPARISON

Forensic Images and Backups

Each row compares the two copies on one point

	Forensic image	Backup copy
What is copied	**Every bit of every usable sector**	Files in folders, or of known types
Deleted files and emails	**Copied** ✓	Not copied ✗
File fragments	**Copied, so they can be recovered** ✓	Not copied ✗
Also called	**A bit-stream image or copy**	A backup

Only a forensic image keeps deleted data and fragments, so examiners image the whole drive.

PART 2 OF 3

Opening a Case

From a blank tool to a loaded forensic image

02

IN THIS PART YOU WILL LEARN TO

- Create a new case in Autopsy
- Add a forensic image as a data source

KEY TERMS

Case **Data source** **Ingest module**

PART 2 · PROCESS

Opening a Case in Autopsy

Four steps, from the Welcome window to a loaded image

1

New case

Case name, base directory, single-user type.

2

Optional details

Case number and examiner name.

3

Add data source

Disk Image or VM (virtual machine) File.

4

Ingest modules

Keep the defaults; click Next, then Finish.

Data source: the evidence you add to a case: here a forensic image ending in .001. Large images are split into .001, .002 and so on.

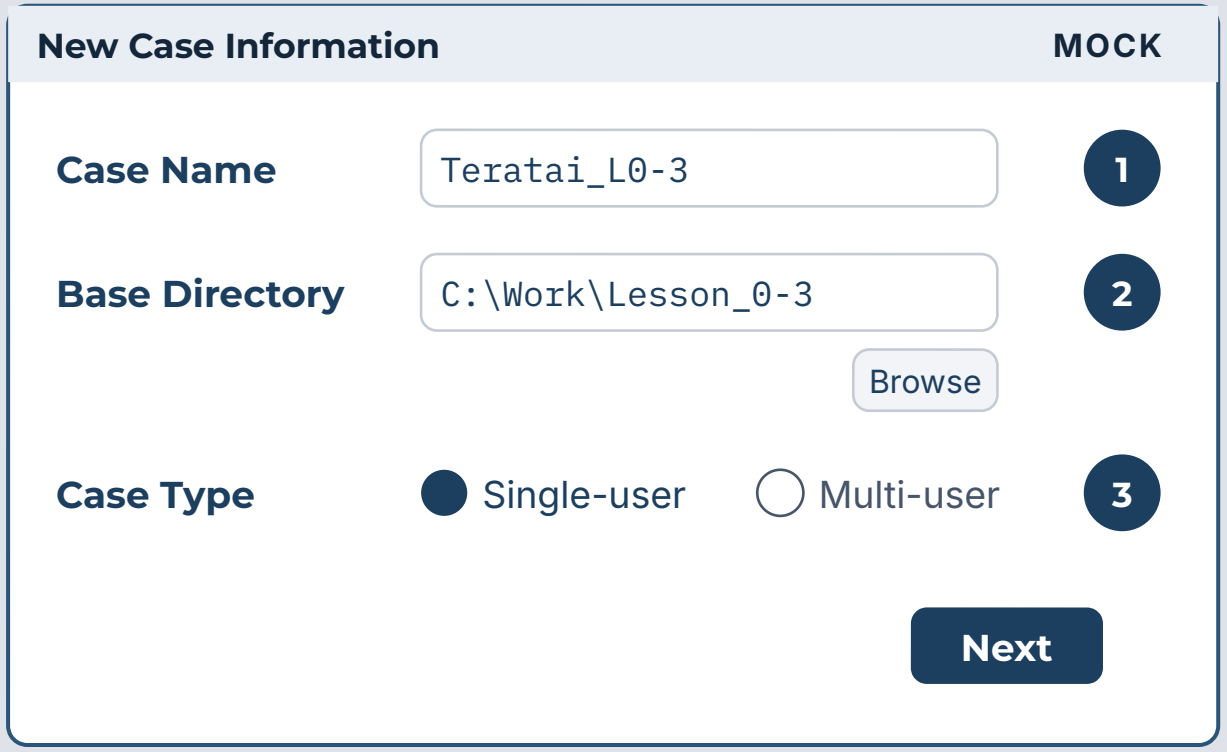
Ingest modules: the parts of Autopsy that process the data source as it is added.

Practise only on the course training image, never on a real device.

PART 2 · WORKED EXAMPLE

Setting Up the Teratai Case

Left: a mock of the first window. Right: what each field is for



1 Case name

Identifies the case; use it in your notes too.

2 Base directory

Your work folder, where the case is saved.

3 Case type

Single-user: one examiner on one computer.

→ Next opens Optional Information: add the case number and your name as examiner.

PART 3 OF 3

Finding Your Way Around

Three panes, and what each one shows

03

IN THIS PART YOU WILL LEARN TO

- Name Autopsy's three main panes
- Find, tag and extract a file

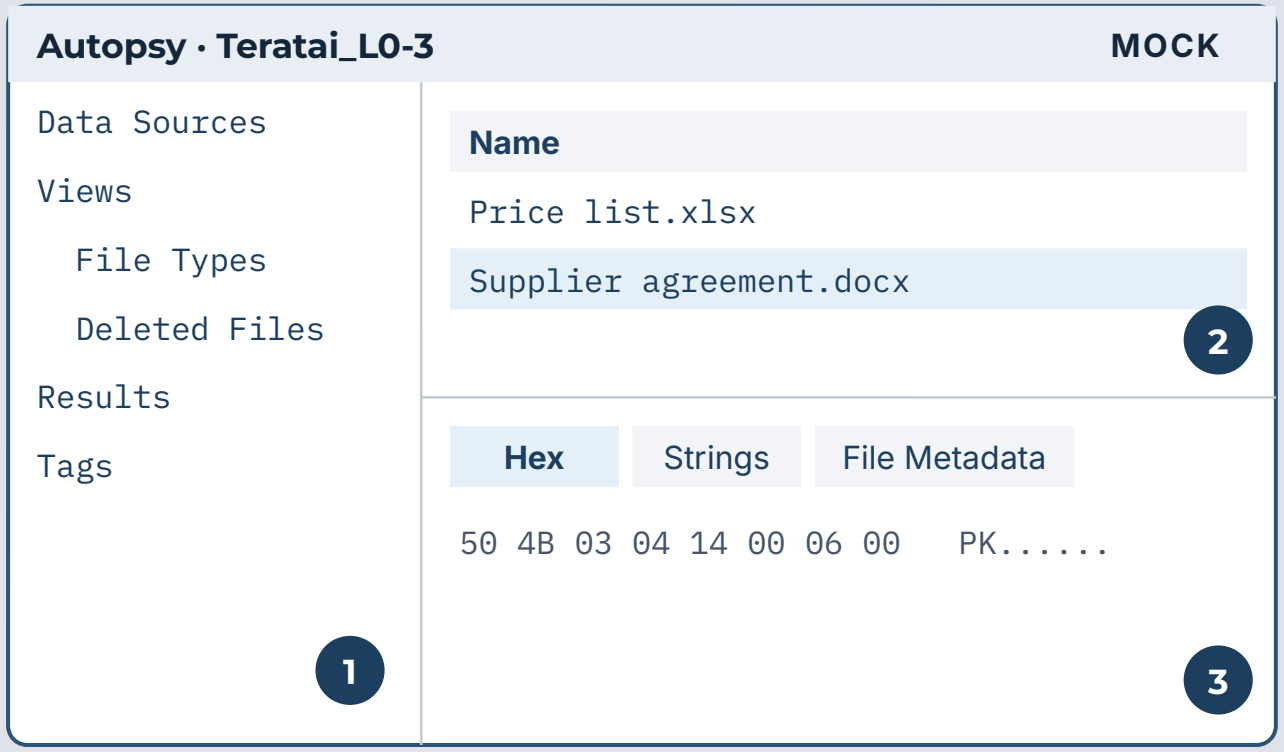
KEY TERMS

Tree Viewer | Result Viewer | Content Viewer | Tag

PART 3 · STRUCTURE

The Autopsy Screen

Left: a mock of the main window. Right: the three panes



1 Tree Viewer, left

The data source in folders and views. Deleted Files sits under Views.

2 Result Viewer, upper right

The files inside whatever is selected in the tree.

3 Content Viewer, lower right

The selected file, as strings of text, hex or metadata.

PART 3 · WORKED EXAMPLE

Finding the Supplier Agreement

Teratai Supplies: the auditors ask whether the image holds a supplier agreement

1 In the Tree Viewer, open the file type

Views → File Types → By Extension → Documents → Office

2 In the Result Viewer, select the file

`Supplier agreement.docx`

Extension: the letters after the dot; .docx is a Word file.

3 In the Content Viewer, read it

The Strings tab shows its text; the Hex tab shows its bytes.

ANSWER

Yes: it is listed under Office documents, found without touching the original drive.

PART 3 · CONCEPT

Working with a File

Three right-click actions you will use from the first lab

Tag and comment

Right-click > **Add File Tag** > **Tag and Comment** > **New Tag**. Name it, e.g. Recovered Office Documents.

Extract a file

Right-click > **Extract File(s)** > **Save**. A copy goes to the Export folder in your case.

Open externally

Right-click > **Open in External Viewer**. The matching program must be installed.

Tag: a label you attach to a file in Autopsy so you can find it again and report on it. Extract saves a copy of the file outside the image.

Where to look: tagged files are listed under Tags in the Tree Viewer.

PART 3 · WORKED EXAMPLE

The Deleted Invoice, Found

Teratai Supplies: the invoice from Lesson 0.2, now in Autopsy



TREE VIEWER (VIEWS)

Deleted Files > All

RESULT VIEWER

`Invoice_0412.xlsx`

YOUR ACTION

Tag, then extract

Why is it still there?

Its entry was marked deleted and its clusters marked free, but no new file has overwritten them yet.

Deleted by the user, still recoverable here: tag it, then extract a copy.

LESSON 0.3 · SUMMARY

Lesson Summary

Three ideas, in the order you met them

1

Before you open a tool

Preserve the original; analyse only a forensic image, never a backup. Hashes prove the copy is exact.

2

Opening a case

New case, optional details, a forensic image as data source, default ingest modules.

3

Finding your way around

Tree, Result and Content Viewers; tag and extract what matters.

Every examination in this course starts the same way: a case, an image, three panes.

UP NEXT

Lesson 0 Summary

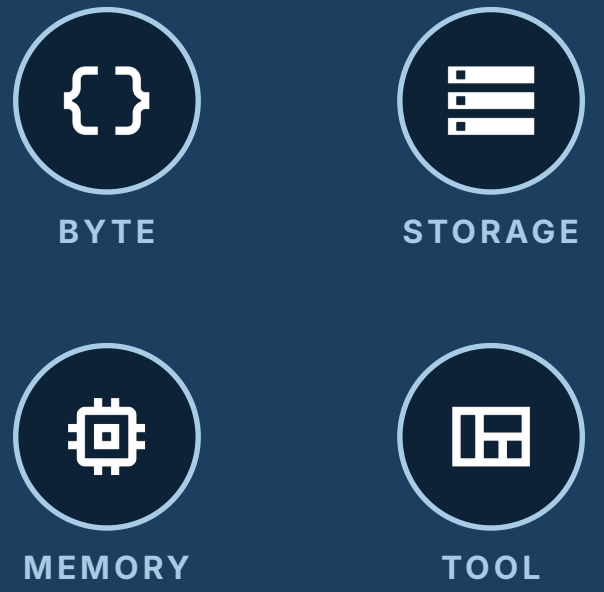
Next, a short review brings Lessons 0.1 to 0.3 together before Lesson 1.

- **Bits, bytes and hexadecimal**
- **Files, storage and memory**
- **Meeting the tools**

DFBOOK DIGITAL FORENSICS · LESSON 0 · SUMMARY

Getting Ready: Lesson Review

Bytes, storage, memory and tools, in one short review



IN THIS LESSON

1. Bits, bytes and hexadecimal
2. Files, storage and memory
3. Meeting the tools

LESSON 0 · REVIEW

Lesson 0 at a Glance

One column for each sub-lesson

0.1 Bits, bytes and hex

1. A byte is eight bits; read as text, ASCII gives it a character.
2. Hex digits run 0 to F; A to F are 10 to 15.
3. 0100 0001 is 65, or 0x41: the letter A.

0.2 Files, storage, memory

1. A file is an entry in the road map.
2. Deleted data can stay until overwritten.
3. RAM is volatile; files on a drive persist.

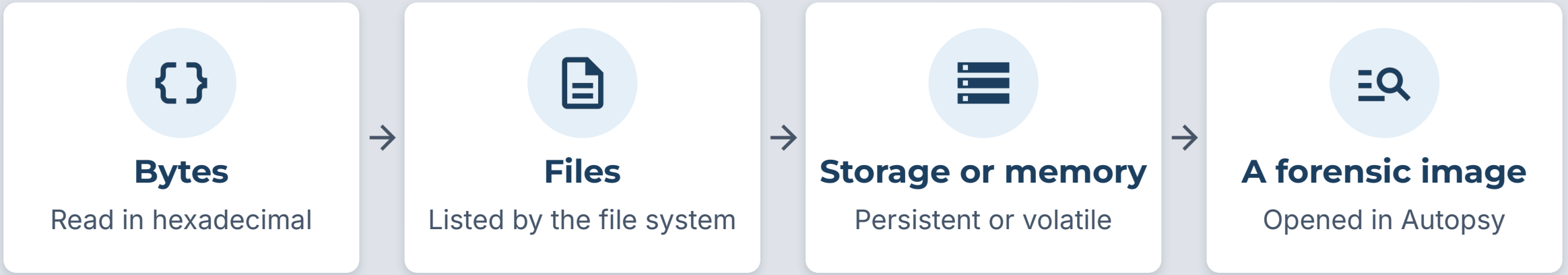
0.3 Meeting the tools

1. Analyse only a forensic image.
2. Add the image to a case as a data source.
3. Tree, Result and Content Viewers.

LESSON 0 · REVIEW

From Bytes to Evidence

How the three sub-lessons fit together



Evidence is bytes: know where they live, and never analyse the original.

UP NEXT

Digital Forensics and the Investigator

Lesson 1 defines the field and its method, and opens the Teratai Supplies case.

- **What digital forensics is**
- **Public and private investigations**
- **A systematic approach to a case**

